

Aditya Kulkarni

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PROFESSIONAL SUMMARY

Engineer specializing in Claude API deployments and agentic systems -- built production multi-model LLM tooling (Claude, GPT-4o, Gemini + MCP) at Isomer AI, delivering MCP server integrations, Temporal.io agentic workflows, and an AI quality evaluation framework benchmarking hallucination rate and latency across 2,000 test cases. Daily Claude Code user with deep familiarity with Anthropic's tooling, model behavior, and deployment patterns. Strong Python and TypeScript across the full stack, 394 commits and 7 production deployments shipping AI-powered features end-to-end.

CORE COMPETENCIES

Claude API + MCP

LLM Deployment + Agents

Eval Frameworks + Prompt Engineering

Python + TypeScript

Agentic Workflows + Temporal

Multi-Model AI (Claude/GPT-4o/Gemini)

REST APIs + AWS + GCP

Docker + CI/CD

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern - Raleigh, NC

- Built production **Claude API + MCP** integrations alongside GPT-4o and Gemini (8+ models, 3 providers); implemented Temporal.io agentic workflow orchestration enabling AI agents to operate within human approval loops end-to-end
- Shipped 10+ backend features across **394 commits and 7 production deployments** -- TypeScript, Node.js, Express, PostgreSQL, Redis, Auth0, Docker, GitHub Actions; designed REST APIs consumed by non-engineer teammates
- Built document processing pipelines (PDF generation, Microsoft Graph API, S3) and prompt-driven workflows; participated in architecture decisions and code reviews as a core member of the engineering team

NC State University Apr 2025 - Present

Graduate Research Assistant - Raleigh, NC

- Deployed a GCP Cloud Run platform enabling 100+ researchers to explore 500GB genomic datasets; applied hyperparameter tuning and time-series analysis, improving ML model accuracy by 12%

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ AI-driven workflows for healthcare claim automation (OCR + NLP + ML); improved accuracy by 25%; automated AWS and GCP deployments with Grafana monitoring

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Led full SDLC for a distributed support ticket analysis system (Kafka + Spark + Hadoop); reduced latency by 20%

PROJECTS

LLM Eval Harness -- AI Quality Evaluation Framework Claude + GPT-4o + Gemini github.com/Aditya-k24/llm-eval-harness

Vendor-neutral eval framework benchmarking Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro on 2,000 frozen test cases across grounded QA, multi-hop QA, and FEVER claim verification. Metrics: ExactMatch, TokenF1, abstention accuracy, grounded hallucination rate, p50/p95 latency, cost/call; statistical significance via paired bootstrap CIs, McNemar test, and Holm correction. Structured prompt templates reusable across model versions and customer deployments.

Python, Anthropic/OpenAI/Google GenAI SDKs, DuckDB, Parquet, Streamlit, Docker

AlgoPulse -- Agentic AI Orchestration Platform Temporal + Kafka + MCP github.com/Aditya-k24/AlgoPulse

Event-driven agentic AI platform with Temporal.io state synchronization and Kafka fault tolerance at 2,000 async inference requests/second; identified API-polling bottleneck mid-build and re-architected to Supabase Edge Functions for streaming AI reasoning. Demonstrates 0-to-1 production rethink under ambiguity -- the same pattern needed when deploying Claude in novel customer environments.

TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka

Glance AI -- Compliance-Ready RAG System RAG + Regulated Domains github.com/Aditya-k24/GlanceAI

RAG system for fintech/healthcare with per-tenant vector retrieval isolation preventing context leakage and LLM hallucinations; deterministic output architecture for compliance-sensitive deployments. Relevant to enterprise LLM deployments in regulated sectors.

Python, LLMs, Vector Databases, RAG, per-tenant isolation

VectorLift -- Semantic Search + ML Pipeline Python + Kafka + FastAPI github.com/Aditya-k24/VectorLift

Kafka-driven indexing pipeline into Elasticsearch and Qdrant; FastAPI search server with per-stage latency tracking; fine-tuned sentence-transformers with in-batch negatives on MS MARCO; evaluated 6 pipeline configurations with NDCG@10, MRR@10, MAP, and bootstrap significance testing.

Python, FastAPI, Kafka, Elasticsearch, Qdrant, PyTorch, Sentence-Transformers, Prometheus, Grafana, Docker

EDUCATION

North Carolina State University -- MS Computer Science Aug 2024 - May 2026

Raleigh, NC | Courses: Artificial Intelligence, Operating Systems, Software Engineering, Algorithms, Data Science

Mumbai University -- BTech Computer Science Aug 2020 - May 2024

CGPA: 9.32/10 | Courses: Data Structures & Algorithms, Machine Learning, Software Engineering, Big Data Infrastructure

SKILLS

AI/LLM: Claude API, MCP, OpenAI, Gemini, LangChain, RAG, PyTorch, Hugging Face, prompt engineering, eval frameworks

Languages: Python, TypeScript, Java, JavaScript, SQL

Backend: Node.js, FastAPI, Express, REST APIs, Kafka, Temporal.io, PostgreSQL, Redis, Docker

Cloud: AWS (S3, automation), GCP (Cloud Run), GitHub Actions CI/CD, Kubernetes